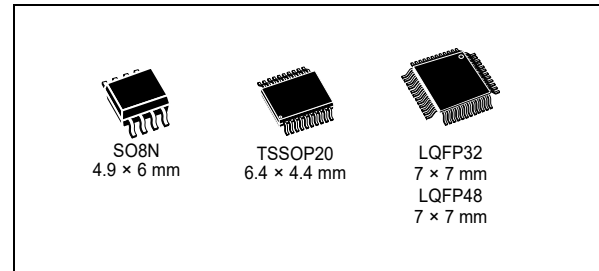


Arm® Cortex®-M0+ 32-bit MCU, up to 64 KB Flash, 8 KB RAM,
2x USART, timers, ADC, comm. I/Fs, 2.0-3.6V

Datasheet - production data

Features

- Core: Arm® 32-bit Cortex®-M0+ CPU, frequency up to 64 MHz
- -40°C to 85°C operating temperature
- Memories
 - Up to 64 Kbytes of Flash memory with protection
 - 8 Kbytes of SRAM with HW parity check
- CRC calculation unit
- Reset and power management
 - Voltage range: 2.0 V to 3.6 V
 - Power-on/Power-down reset (POR/PDR)
 - Low-power modes: Sleep, Stop, Standby
 - V_{BAT} supply for RTC and backup registers
- Clock management
 - 4 to 48 MHz crystal oscillator
 - 32 kHz crystal oscillator with calibration
 - Internal 16 MHz RC with PLL option
 - Internal 32 kHz RC oscillator (±5 %)
- Up to 44 fast I/Os
 - All mappable on external interrupt vectors
 - Multiple 5 V-tolerant I/Os
- 5-channel DMA controller with flexible mapping
- 12-bit, 0.4 µs ADC (up to 16 ext. channels)
 - Up to 16-bit with hardware oversampling
 - Conversion range: 0 to 3.6V
- 8 timers: 16-bit for advanced motor control, four 16-bit general-purpose, two watchdogs, SysTick timer
- Calendar RTC with alarm and periodic wakeup from Stop/Standby



- Communication interfaces
 - Two I²C-bus interfaces supporting Fast-mode Plus (1 Mbit/s) with extra current sink, one supporting SMBus/PMBus and wakeup from Stop mode
 - Two USARTs with master/slave synchronous SPI; one supporting ISO7816 interface, LIN, IrDA capability, auto baud rate detection and wakeup feature
 - Two SPIs (32 Mbit/s) with 4- to 16-bit programmable bitframe, one multiplexed with I²S interface
- Development support: serial wire debug (SWD)
- All packages ECOPACK 2 compliant

Table 1. Device summary

Reference	Part number
STM32G030x6	STM32G030C6, STM32G030F6, STM32G030J6, STM32G030K6
STM32G030x8	STM32G030C8, STM32G030K8

List of tables

Table 1.	Device summary	1
Table 2.	STM32G030x6/x8 family device features and peripheral counts	10
Table 3.	Access status versus readout protection level and execution modes.	13
Table 4.	Interconnect of STM32G030x6/x8 peripherals	17
Table 5.	Temperature sensor calibration values.	22
Table 6.	Internal voltage reference calibration values	23
Table 7.	Timer feature comparison.	23
Table 8.	I ² C implementation	26
Table 9.	USART implementation	27
Table 10.	SPI/I2S implementation	28
Table 11.	Terms and symbols used in Table 12	30
Table 12.	Pin assignment and description	31
Table 13.	Port A alternate function mapping	35
Table 14.	Port B alternate function mapping	36
Table 15.	Port C alternate function mapping	37
Table 16.	Port D alternate function mapping	37
Table 17.	Port F alternate function mapping.	37
Table 18.	Voltage characteristics	40
Table 19.	Current characteristics	40
Table 20.	Thermal characteristics.	41
Table 21.	General operating conditions	41
Table 22.	Operating conditions at power-up / power-down	41
Table 23.	Embedded reset and power control block characteristics.	42
Table 24.	Embedded internal voltage reference.	42
Table 25.	Current consumption in Run and Low-power run modes at different die temperatures	44
Table 26.	Current consumption in Sleep and Low-power sleep modes	45
Table 27.	Current consumption in Stop 0 mode	45
Table 28.	Current consumption in Stop 1 mode	46
Table 29.	Current consumption in Standby mode	46
Table 30.	Current consumption in VBAT mode	46
Table 31.	Current consumption of peripherals	48
Table 32.	Low-power mode wakeup times	49
Table 33.	Regulator mode transition times	50
Table 34.	High-speed external user clock characteristics.	50
Table 35.	Low-speed external user clock characteristics	51
Table 36.	HSE oscillator characteristics	51
Table 37.	LSE oscillator characteristics ($f_{LSE} = 32.768$ kHz)	53
Table 38.	HSI16 oscillator characteristics.	54
Table 39.	LSI oscillator characteristics	55
Table 40.	PLL characteristics	55
Table 41.	Flash memory characteristics	56
Table 42.	Flash memory endurance and data retention	56
Table 43.	EMS characteristics	57
Table 44.	EMI characteristics	58
Table 45.	ESD absolute maximum ratings	58
Table 46.	Electrical sensitivity.	59
Table 47.	I/O current injection susceptibility	59